

THIWANKA LAKSHAN

Full Stack Developer

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PROFILE SUMMARY

Full-Stack Developer with a robust foundation in software engineering principles, agile methodologies, and modern web architectures. Proficient in architecting scalable JavaScript environments using **React.js, Vue.js, Node.js, and Express.js**, with proven hands-on experience building **IoT-driven systems, RESTful APIs, and relational databases (PostgreSQL/MySQL)**. Strong problem-solving mindset focused on writing clean, tested code and delivering efficient, user-centric solutions.

EDUCATION

University of Vavuniya

Bachelor of Information and Communication Technology Honours (BICT Hons) Degree

June 2023 – Present

PROJECTS

MooMap – Real-Time Cattle Monitoring using IoT Smart Collars & LoRa

([View](#))

Technologies: Node.js, Express.js, PostgreSQL, REST APIs

- MooMap is an IoT-based livestock monitoring system that enables real-time cattle tracking using GPS-enabled smart collars and LoRa communication, delivering live data to web and mobile clients.
- Designed and optimized relational PostgreSQL database schemas, indexing live tracking metrics to reduce query latency for real-time client requests.
- Developed scalable server-side logic for real-time data processing, alerting, and secure delivery to web and mobile clients.

FloodSense – Real-Time Flood Prediction and Early Warning System

([View](#))

Technologies: Node.js, Express.js, React, Vue.js

- FloodSense is a real-time flood monitoring and early warning system for Sri Lanka's Western Province, providing automated email alerts with 2–24 hour evacuation lead time across 20+ locations in Colombo, Gampaha, and Kalutara districts.
- Engineered a 5-factor rule-based algorithm achieving 85% prediction accuracy for short-term flood forecasts.
- Architected a Full-Stack application utilizing a Node.js/Express REST API backend coupled with React/Vue.js architectures to manage high-frequency data collection.

TECHNICAL SKILLS

Programming Languages: JavaScript, Python, Java, C++

Frontend: React.js, Vue.js, HTML5, CSS3

Backend: Node.js, Express.js, RESTful API, Solidity, JWT Authentication

Databases: PostgreSQL, MySQL, MongoDB

Tools & Technologies: Docker, Kubernetes, GitHub, Vercel, Render, Postman, VS Code

Core Concepts: OOP, Data Structures & Algorithms, SDLC, Agile

CERTIFICATIONS

• Blockchain Specialization – University of Buffalo	July 2026	(View)
• Developing Back-End Apps with Node.js and Express – IBM	October 2025	(View)
• Developing Front-End Apps with React – IBM	December 2025	(View)
• Getting Started with Git and GitHub – IBM	September 2025	(View)
• Introduction to Containers w/ Docker, Kubernetes & OpenShift – IBM	June 2026	(View)
• Python for Data Science, AI & Development – IBM	April 2026	(View)

EXPERIENCE

Vice Chair (Volunteering)

IEEE Computer Society, University of Vavuniya ·

October 2025 – Present

- Led as project Vice Chair for IEEE Andaharaya 2025, spearheading a nationwide program to increase IEEE membership awareness and engagement.
- Represented the IEEE Student Branch of the University of Vavuniya at IEEE LeaderConn 2026, gaining leadership and team management experience.

Team Lead (Volunteering)

IEEE Student Branch, University of Vavuniya ·

April 2025 – Present

- Directed Program and Logistics for IEEE Day 2025, managing event execution and developing leadership and team coordination skills.
- Captained Team BugBusters9588 at IEEEExtreme 19.0 (24-hour global coding contest), competing against 19,000+ participants and applying problem-solving, teamwork, and decision-making under pressure.

REFEREES

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